Development tools for biomimetic fish robots

"Make a full report of the limitations and possible scopes for development and research about any and all tools helping us do so for example for PINNs - NVIDIA Omniverse; CFD - OpenFOAM literature; Soft Robotics - SpineWave etc." *Generated by LeapSpace*

Full report: limitations and development scope for enabling tools

The evidence points to a **toolchain problem, not a single-model problem**: progress in biomimetic underwater robotics depends on combining **PINNs / physics-informed learning**, **CFD and FSI solvers such as OpenFOAM and Fluent**, and **soft-robotics simulation/control frameworks such as SOFA**, but each layer still has clear bottlenecks in accuracy, scalability, validation, and real-world deployment [22] [23] [24] [97] [98] [99] [100] [101] [102].

What each tool class is good for, and where it breaks down

Tool class	What it already enables	Main limitations	Development / research scope
PINNs and physics-informed learning	Forward and inverse modeling with embedded governing laws; useful for fluid dynamics, continuum robots, and control with limited data [97] [100] [103] [104] [105] [106] [107] [108]	High computational cost, convergence instability, limited multiscale generalization, difficulty with turbulence, nonlinear forces, and global constraints [100] [109] [110] [111] [112]	Better loss balancing, constrained architectures, serial / modular PINNs, hybrid physics-data models, and standardized benchmark sets for fluid and robot dynamics [100] [101] [103] [106] [110] [111]
CFD / FSI solvers, especially OpenFOAM	Hydrodynamic analysis of biomimetic fish, turbulence evaluation, wake visualization, and design comparison; OpenFOAM is used to build validated open setups for fish-like bodies [98] [102] [113] [114] [115] [116]	Major gaps remain in open reference geometries, calibration/validation data, suitable turbulence modeling, boundary-layer treatment, and computational expense [102] [113] [117]	Open benchmark geometries, better near-wall models, validated fish-shaped reference cases, and coupling CFD with ML surrogates for faster optimization [102] [113] [118] [119]

Tool class	What it already enables	Main limitations	Development / research scope
Soft robotics simulation and design tools	Modeling of continuum bodies, compliant fins, variable stiffness, pneumatic actuation, and underwater adaptation; SOFA is used where rigid simulators are insufficient [99] [112] [120] [121] [122] [123] [124]	Soft underwater robots are hard to model and control because of high DOF, water-tightness problems, complex fluid coupling, sparse sensing, material nonlinearity, and pressure sensitivity [99] [101] [123] [125] [126]	Rapid fabrication, programmable stiffness, dense sensing, proprioception, co-design of body and controller, and better real-time model reduction for soft underwater motion [99] [105] [112] [127] [128] [129]
Control / autonomy stacks, including ROS-like integration	Autonomy, obstacle avoidance, DRL, CPG control, sim-to-real transfer, and modular robot operation [130] [131] [132] [133] [134] [135]	Existing biomimetic fish robots are still mostly prototypes with relatively simple functions; remote sensing/control and robust navigation remain underdeveloped [127] [131] [134]	Closed-loop, sensor-rich control integrated with hydrodynamic surrogates, better sim-to-real pipelines, and middleware-standardized robot stacks for field deployment [104] [112] [131] [133]

Key limitations by theme

1) PINNs: promising, but still not a turnkey fluid solver

PINNs are repeatedly described as attractive because they embed physical laws directly into the loss function and can solve forward and inverse problems with sparse data [97] [100] [109]. The main blockers are **training instability, high cost, poor turbulence handling, and weak multiscale generalization** [100] [109]. For robotics applications, the literature also shows that traditional PINNs struggle when nonlinear forces are unclear or when the system has nonconservative effects [104] [111].
Research scope: constrained PINN variants, hybrid PINN-ROM frameworks, serial/multi-network formulations, and benchmark suites tied to robot-relevant fluid problems [101] [103] [106] [110] [111].

2) OpenFOAM / CFD: strong for hydrodynamics, weak without good validation

OpenFOAM-based work demonstrates useful hydrodynamic analysis for biomimetic robot fish, including dynamic meshes and comparative thrust/drag studies [98] [113] [114] [115]. But the CFD literature also identifies **missing open geometries, incomplete calibration data, uncertain turbulence models, and boundary-layer trade-offs** [102] [113]. A separate CFD/FSI paper also notes that mesh refinement and coupled-field computation are central challenges [117].
Research scope: validated open fish-like benchmarks, standardized calibration datasets, better wall modeling, and surrogate-assisted CFD to reduce runtime [102] [113] [118] [119].

3) Soft robotics: the hardware is flexible, but the ecosystem is not yet mature

Soft underwater robots promise adaptability, safety, and better environmental interaction, but they are difficult to model and control because of **high degrees of freedom, coupling with water, waterproofing, sparse sensing, and nonlinear material behavior** [99] [125] [127]. The literature also shows that soft robot fish often still rely on basic tasks like depth and heading control [127].
Research scope: programmable stiffness, better proprioception and lateral-line-like sensing, co-design of actuation and control, and more robust underwater materials/fabrication [99] [105] [112] [128] [129].

4) Control and autonomy: better algorithms exist than integrated systems

Control research already includes CPGs, DRL, PID, and sim-to-real strategies for robotic fish [133] [134] [136]. However, the broader review evidence says biomimetic fish robots are still **mostly prototypes with relatively simple functions**, and remote sensing/control remains a major gap [127] [131].
Research scope: integrate physical modeling, sensing, and control into a single stack; use physics-informed surrogates online; and move from isolated demonstrations to full mission-level autonomy [104] [112] [133].

Best-supported development directions

1. **Hybrid digital twins for underwater robots** combining CFD/OpenFOAM, PINNs, and soft-body simulation to reduce the cost of design-space exploration [98] [100] [101] [102] [119].
2. **Benchmark-driven soft robot design**, where new materials, variable stiffness, and sensing are evaluated against shared hydrodynamic and control tasks [99] [123] [127] [128].
3. **Surrogate and reduced-order models** for long-horizon prediction and real-time control, especially where full CFD is too expensive [111] [112] [118] [119].
4. **Physics-informed control loops** that merge model-based control, data-driven learning, and physical constraints for robustness under uncertainty [104] [105] [129] [133].
5. **Standardized validation for low-disturbance operation**, including wake, drag, noise, and ecological-response measurements, because current evidence still emphasizes hydrodynamics more than field disturbance metrics [102] [113] [137] [138].

Bottom line

The clearest research opportunity is **not just better fish robots, but better integration infrastructure**: a pipeline that links **validated CFD/OpenFOAM**, **scalable PINNs or physics-informed surrogates**, **soft-body simulation and sensing**, and **field-ready autonomy**. The literature supports each piece separately, but it also shows that their **joint, validated deployment for low-disturbance monitoring remains largely open** [22] [23] [24] [97] [98] [99] [100] [102] [133].

Confidence level: **High** — the main limitations and research directions are directly and consistently stated across multiple reviews, simulation studies, and control papers, with strong agreement between the prior synthesis and the newly retrieved evidence [97] [99] [100] [102] [104] [113].

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